Wrapper Classes in Java

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Library classes Introduction to wrapper classes, methods of wrapper class and their usage with respect to numeric and character data types. Autoboxing and Unboxing in wrapper classes. Class as a composite type, distinction between primitive data type and composite data type or class types. Class may be considered as a new data type created by the user, that has its own functionality. The distinction between primitive and composite types should be discussed through examples. Show how classes allow user defined types in programs. All primitive types have corresponding class wrappers. Introduce Autoboxing and Unboxing with their definition and simple examples.

The following methods are to be covered:
int parseInt(String s), long parseLong(String s), float parseFloat(String s), double parseDouble(String s), boolean isDigit(char ch), boolean isLetter(char ch), boolean isLetterOrDigit(char ch), boolean isLowerCase(char ch), boolean isUpperCase(char ch), boolean isWhitespace(char ch), char toLowerCase (char ch) char toUpperCase(char ch)

Boxing, Autoboxing and Unboxing in Java

In Java, boxing, auto boxing and unboxing are concept that explain the conversion between primitive data types and their corresponding wrapper classes. Java provides. These features to make it easier to work with primitive values as objects.

Understanding primitive data types and a wrapper classes

Before understanding boxing and unboxing, we must understand primitive data types and there is corresponding wrapper classes.

Java has eight primitive data types. Each primitive data type has a corresponding wrapper class available in java.lang package.

Primitive Data Type	Wrapper class
byte	Byte
short	Short
int	Integer
long	Long
float	Float
double	Double
char	Character
boolean	Boolean

```
int x = 100;
x
```

← primitive type

```
x
100   (int)
Integer i = 200;
i
<object reference>   (Integer)
System.out.print(i);
i = x;
i
<object reference>   (Integer)
System.out.print("\n"+i);
```

Integer is a wrapper class.
The Integer class wraps an integer value inside an object.
This is called boxing.

What is Boxing in Java?
Boxing is the process of converting a primitive data type into its corresponding wrapper class object.
In simple words, converting a primitive value into an object is called **Boxing.**

The main purpose of Boxing is we get multiple methods to manipulate primitive data.

What is Autoboxing in Java?
Auto boxing is the automatic conversion of a primitive data type into its corresponding wrapper class object by the Java compiler.
In simple words, when Java automatically converts a primitive value into an object, it is called auto boxing.

```
Integer x = 100;
```

What is unboxing in Java?
Unboxing is the process of converting a wrapper class object into its corresponding primitive data type.

In simple words, converting an object into a primitive value is called unboxing.

```
Integer x = 100; //boxing
int y = x; //Unboxing
```

What is automatic unboxing in Java?
Automatic unboxing is the automatic conversion of wrapper class object into its corresponding primitive data type by the Java compiler.

Difference between boxing, auto boxing and unboxing?

Concept	Meaning	Conversion

Boxing	Explicit conversion of primitive into object.	int->Integer
Autoboxing	Automatic conversion of primitive into object.	int->Integer
Unboxing	Explicit conversion of object into primitive.	Integer->int
Automatic unboxing	Automatic conversion of object into primitive.	Integer->int

Remember these statement:

```
//Boxing
Integer obj1 = Integer.valueOf(100);

//Autoboxing
Integer obj2 = 100;

//Unboxing
int a = obj1.intValue();

//Automatic Unboxing
int b = obj2;
```